Elizabeth Elizondo

Project: Cave LLC Shippers Co. Analysis
Date: 05/2023

Description of Problem Statement

Cave LLC Shippers Co. is seeking growth for future development. In order to do this, analysis of data was pulled from the beginning of 2021 to the middle of the 4th quarter in 2022 Due to Cave LLC being the first container manufacturer of its kind, further analysis is required to determine if sales by container composition type (material). Depending on what type of information the data reveals, the next step is to determine our company should:

Data Description

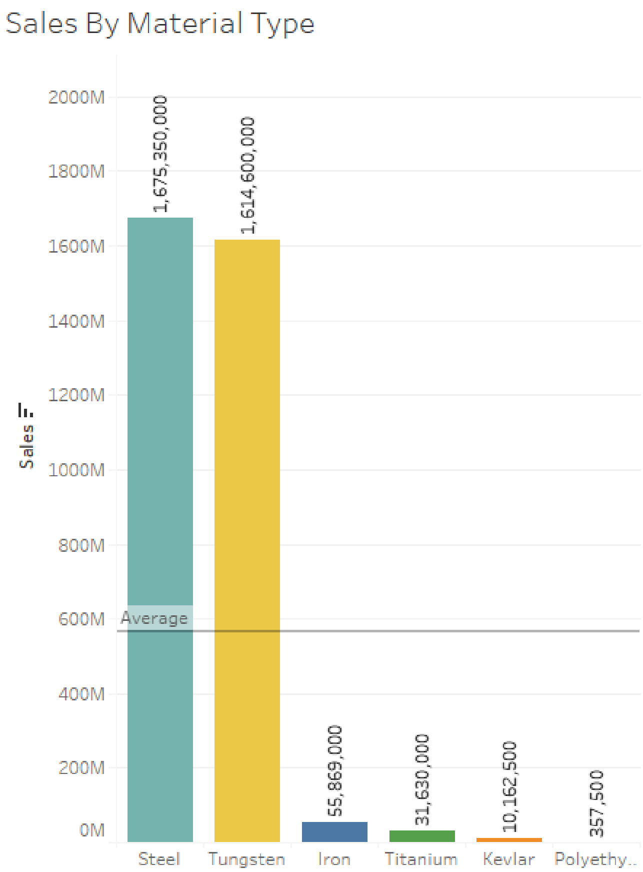
Before Cleaning: How much data is there? 9215 rows, 14 Columns
After Cleaning: Historical data covers year 2022 broken down into quarters.
6971 rows of data, each row includes: OrderID, Origin Port, Carrier, ServiceLevel, ShipAheadDayCount, ShipLateDayCount, Customer, ProductID, PlantCode, DesitnationPort, UnitQuantity, Weight, Material, Retail, Sales, Orderdate.

Analysis Tasks

1.) Continue to manufacture the same type of shipper containers
2.) Determine which shippers have the most sales
3.) Customer base trends
4.) Statistical Analysis
5.) Forecast based on sales

Exploratory Data Analysis

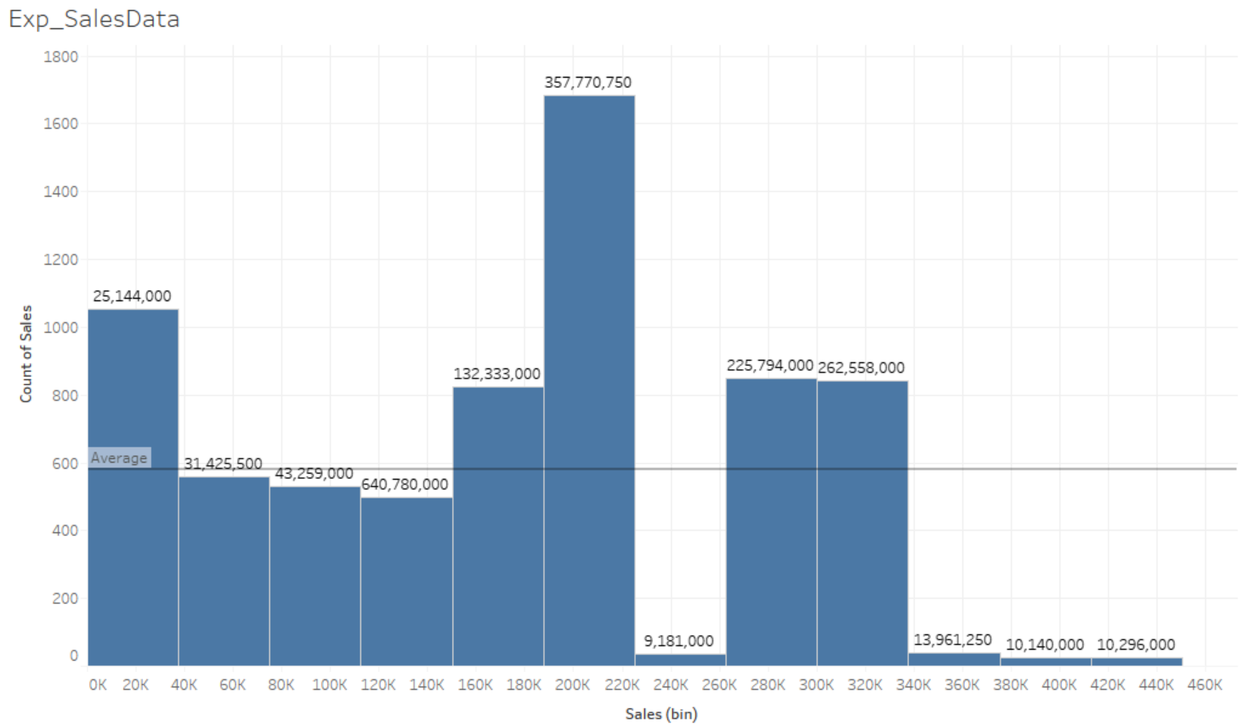
Question to be answered: Which type of shipper material generated the most cost?
Assumption: Ideally, steel has been the main source for shipper composition, we would expect to see this type as highest as most customers will purchase a item they are familiar with in the industry.



Insights:

- Steel and Tungsten among best sellers
- Average sales for all material types are roughly 600 million
- Polyethylene lowest sold material type

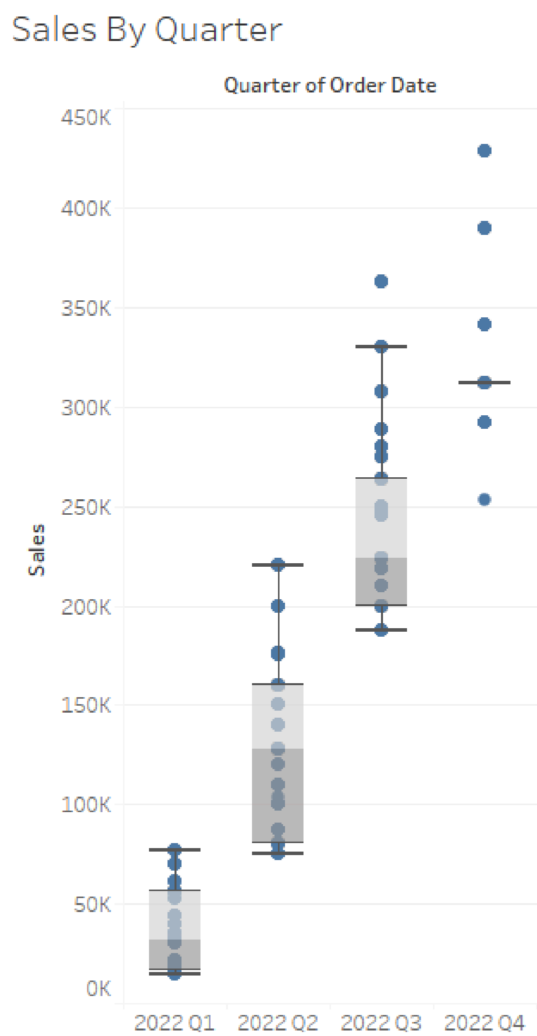
Next, we seek to explore what the company's sales data looks like from a high level to better understand frequency and range of values using a histogram chart.
Question to be answered: How many sales were generated that brought in what dollar amount?
Assumption: Ideally, the larger the count of sale, the higher the sales bin will be.



Insights:

- Sales lows 9,181,000 with 37 count of sales, versus 9,181,000 with only 24 count of sales made 10,296,000 - This indicates less sales generated a higher yield return value for a specific product that may have cost more to the consumer.
- Sales high was 1,685 count of sales yielded $357,770,750 compared to its 2nd highest sales count at 1,054 returning sales dollar amount of $25,144,000.

Next, we seek to explore what the company's sales data looks like by using a box plot in order to demonstrate the five-number summary of the sales data set: including the minimum score, first (lower) quartile, median, third (upper) quartile, and maximum score.
Question to be answered: What does the sales by quarter tell us?
Assumption: No assumptions



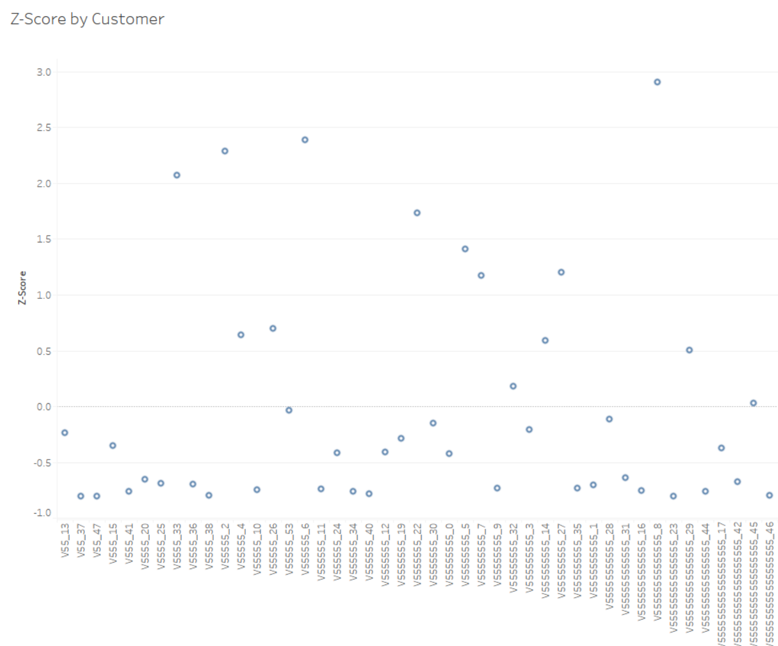
Insights:

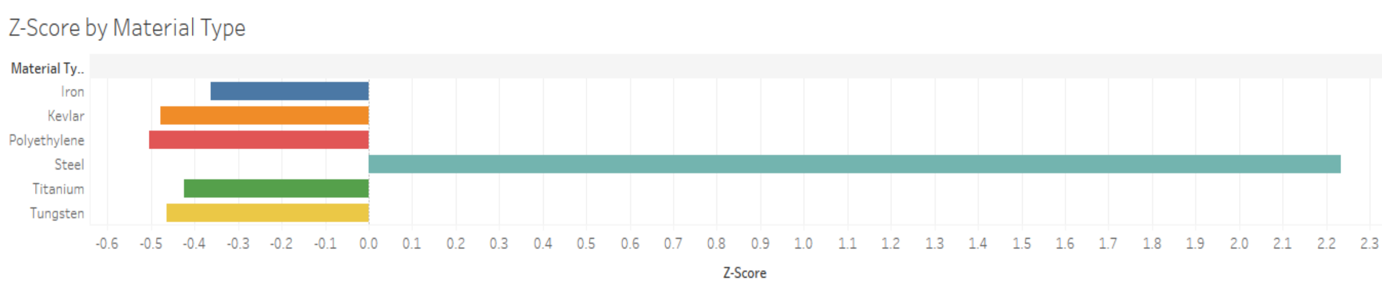
- Q1: The interquartile range (IQR) allows is positively skewed, does contain some outliers however data is less dispersed to due starting sales frequency at a low.
- Q2: IQR has normal distribution with a few outliers which have wider dispersion with a greater increase in sales.
- Q3: IQR is positively skewed similar Q1 with the most outliers, an indicator that sales are more inconsistent this quarter.
- Q4: Data points only pinpointing few sales data generated for Q4

Z-Score

In order to identify observations with unusually large or unusually small values ie. outliers that may need to be further investigated, therefore the z-score will help determine that. Following empirical rule all data values should fall within 3 standard deviations of mean, therefore any values with a z-score less than -3 or greater +3 is considered an outlier. However, as a general rule, a z-score with lower than a -1.96 or higher of a 1.96 are considered an anomaly.

Question to be answered: Where does the z-score fall within material type and customer orders and frequency of order date?
Assumption: Z-score will fall within the permissible limitations of -3 or +3.
To calculate a Z-Score the population mean and population standard deviation is calculated by the following:
Average Sales: WINDOW_AVG(SUM([Sales]))
Standard Deviation: WINDOW_STDEVP(SUM([Sales]))
Z-Score: (SUM([Sales]) - [Average Sales]) / [STDEVP Sales]

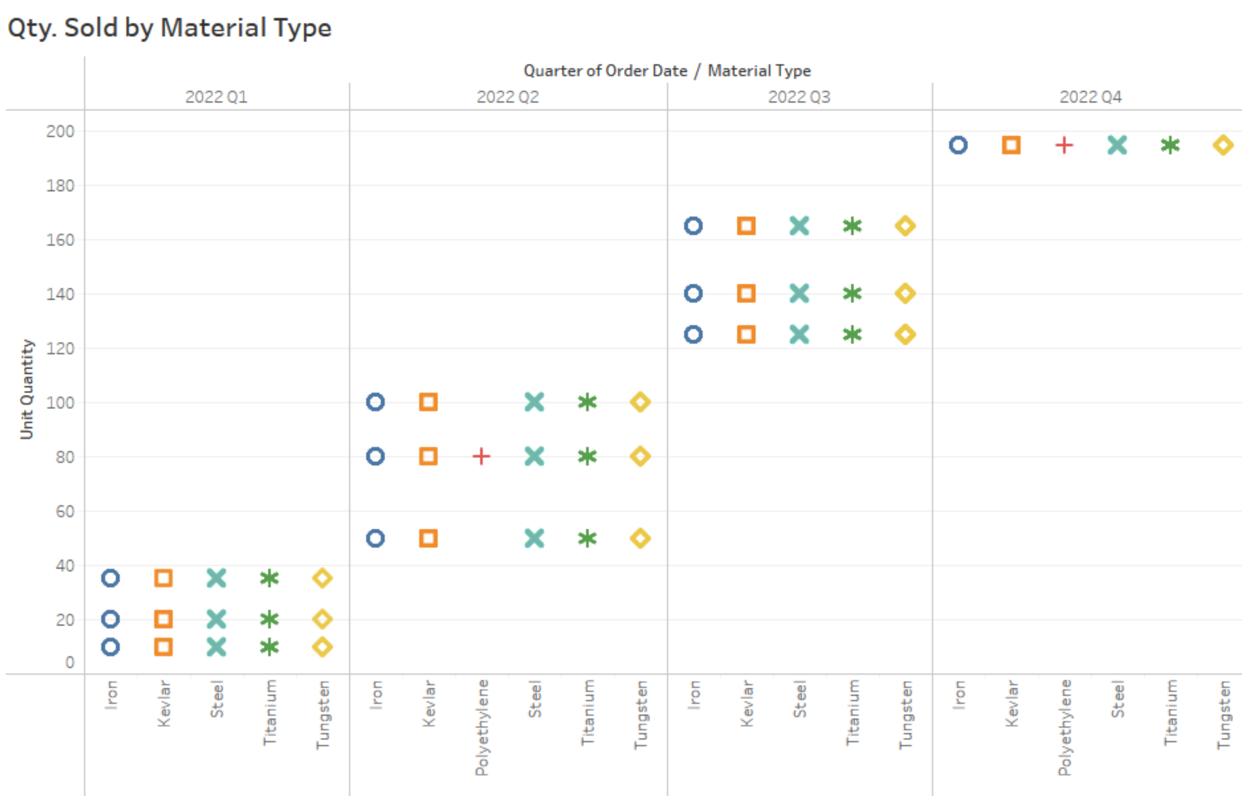




Insights:

- All shipper material types fall less than or near -0.5 whereas Steel falls slightly over 2.2 which indicates that it an outlier with steel as it is the material most sold out of all categories
- Order date frequency was observed for its z-score and it was determined that quarter Contains an anomaly at 4.027 for order no. 1447371268 which indicates a greater frequency sold than normal, this could be attributed to the approaching holiday season and needs further investigation.
- Sales by customer were all consistent within the +3 z-score permissible limitation.

Question to be answered: What does the material quantity by quarter tell us?
Assumption: Sales would increase in range progressively each quarter.

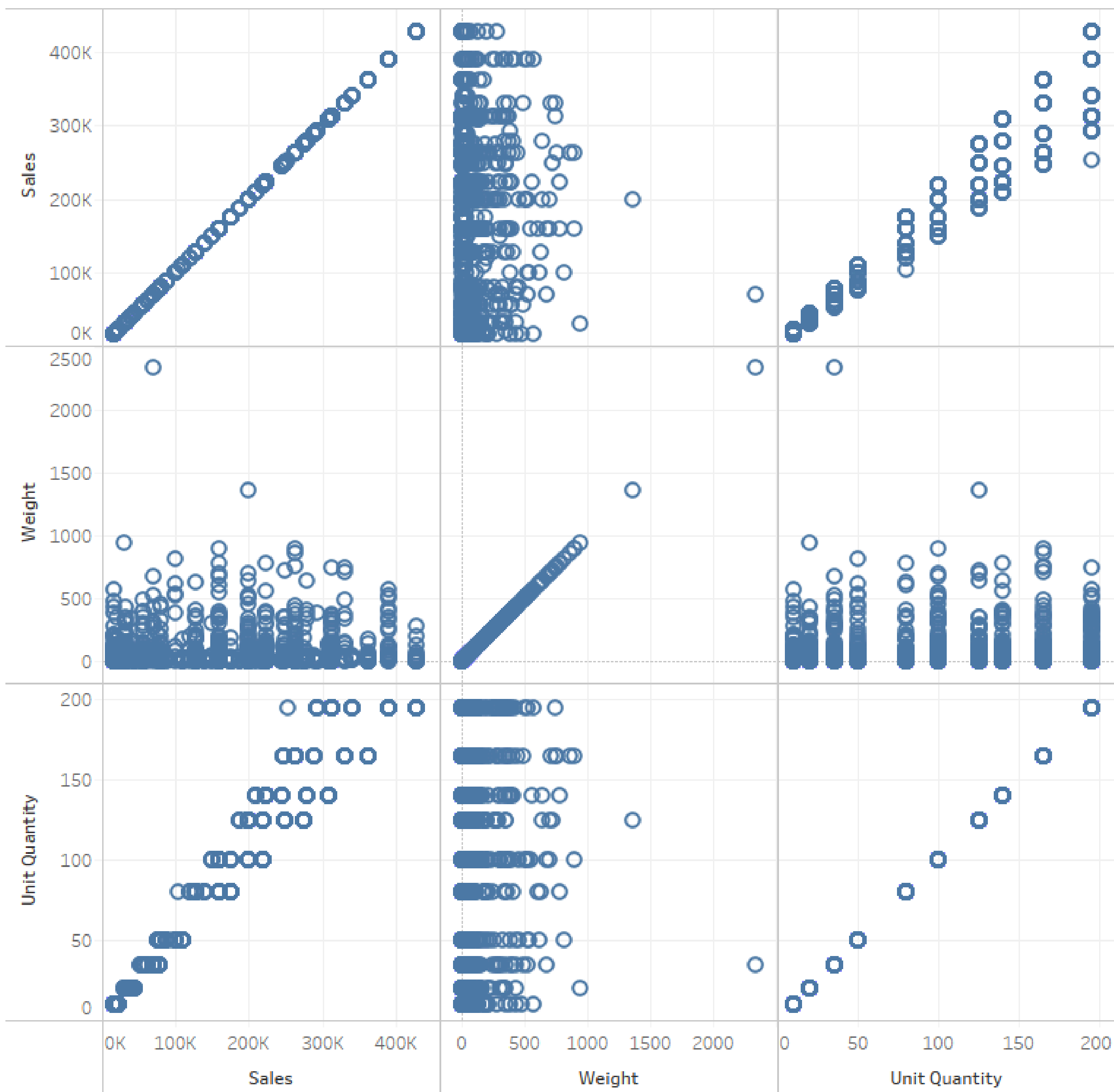


Insights:

- Sales progress in a positive trend within each material category
- Polyethylene shows no units sold in Q1 and Q3

Question to be answered: What does quantity sold, weight, and sales tell us from a scatter-chart matrix?
Assumption: No assumptions.

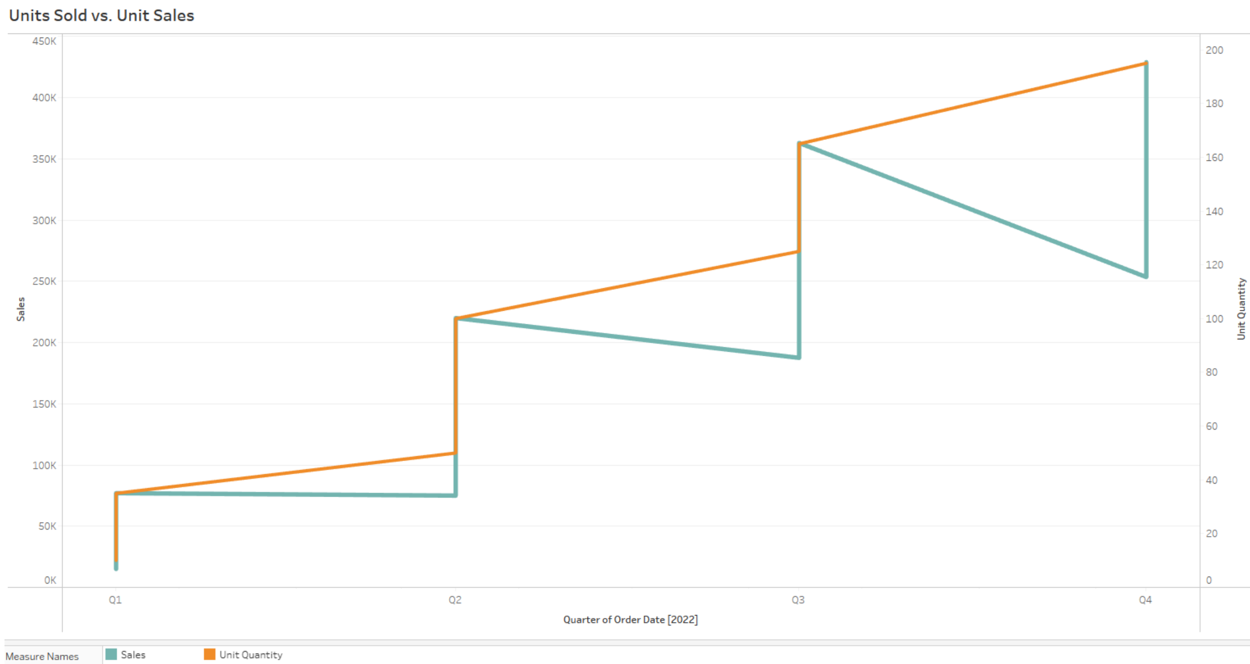
Scatter-Chart Matrix for Sales Data



Insights:

- Sales by weight appear to be approximately under 1000 lbs. However a couple of outliers are present.
- Unit quantity trend positive along with sales increasing
- Unit quantity have an even spread up to 200 range, despite weight fluctuations under 1000 lbs.

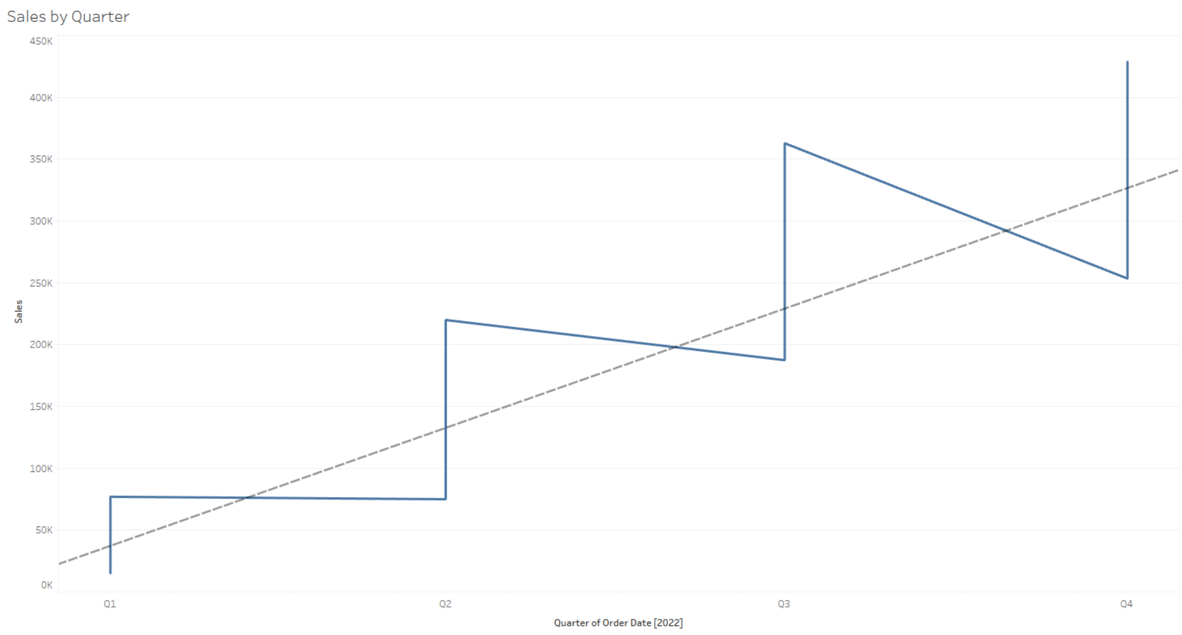
Question to be answered: How does units sold impact sales?
Assumption: The greater the units, the greater the sales generated.



Insights:

- Q1 into Q2 have steady sales, but later decline at the end of each quarter before peaking at the beginning of the next. This would indicate that most of our buyers budget to spend at the beginning of each quarter and purchase less units as the quarter completes to its end.
- Units (orange line) show a consistent climb in increased numbers, where the sales drop as the units increase. This would indicate that there is a discount applied with the greater number in quantity purchased due to the decline in sales.

Statistical Models



Trend Lines Model

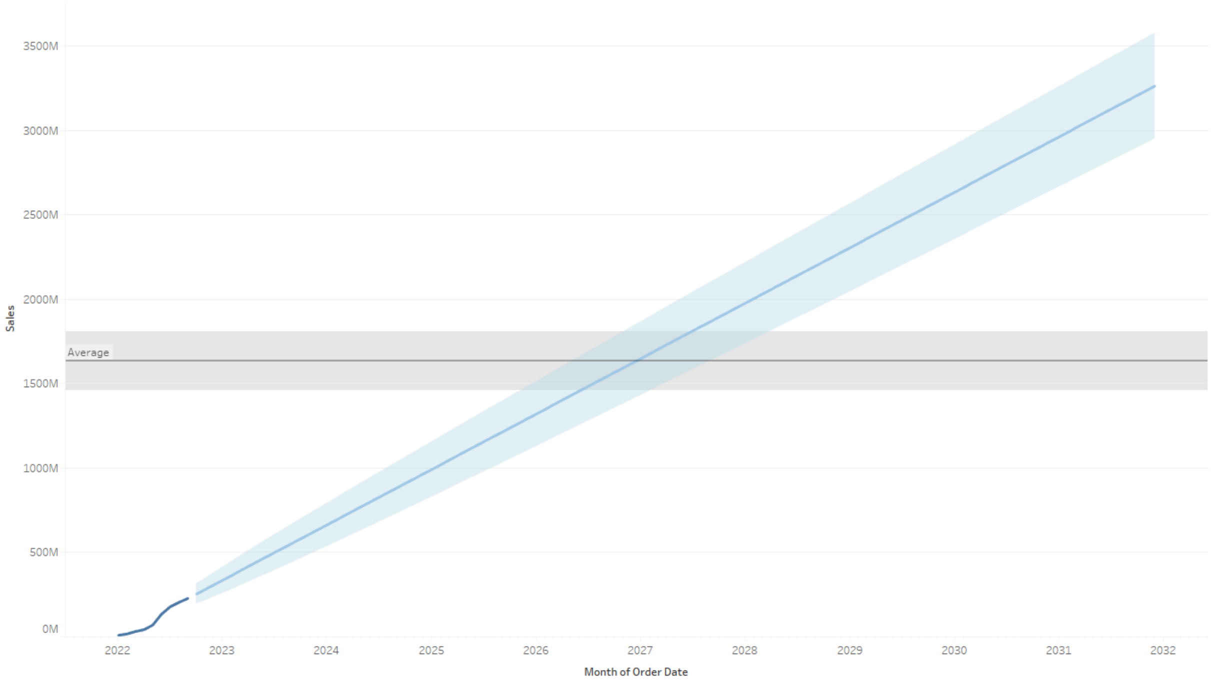
A linear trend model is computed for Sales given Quarter of Order Date. The model may be significant at $p <= 0.05$.

Model formula:	(Quarter of Order Date + intercept)
Number of modeled observations:	6971
Number of filtered observations:	0
Model degrees of freedom:	2
Residual degrees of freedom (DF):	6969
SSE (sum squared error):	6.0835e+12
MSE (mean squared error):	8.72937e+08
R-Squared:	0.909871
Standard error:	29545.5
p-value (significance):	< 0.0001

Individual trend lines:

Panes		Line		Coefficients				
Row	**Column**	**p-value**	**DF**	**Term**	**Value**	**StdErr**	**t-value**	**p-value**
Sales	Quarter of Order Date	< 0.0001	6969	Quarter of Order Date	1060.32	3.99755	265.242	< 0.0001
				intercept	-4.72131e+07	178642	-264.289	< 0.0001

10-Year Sales Forecast



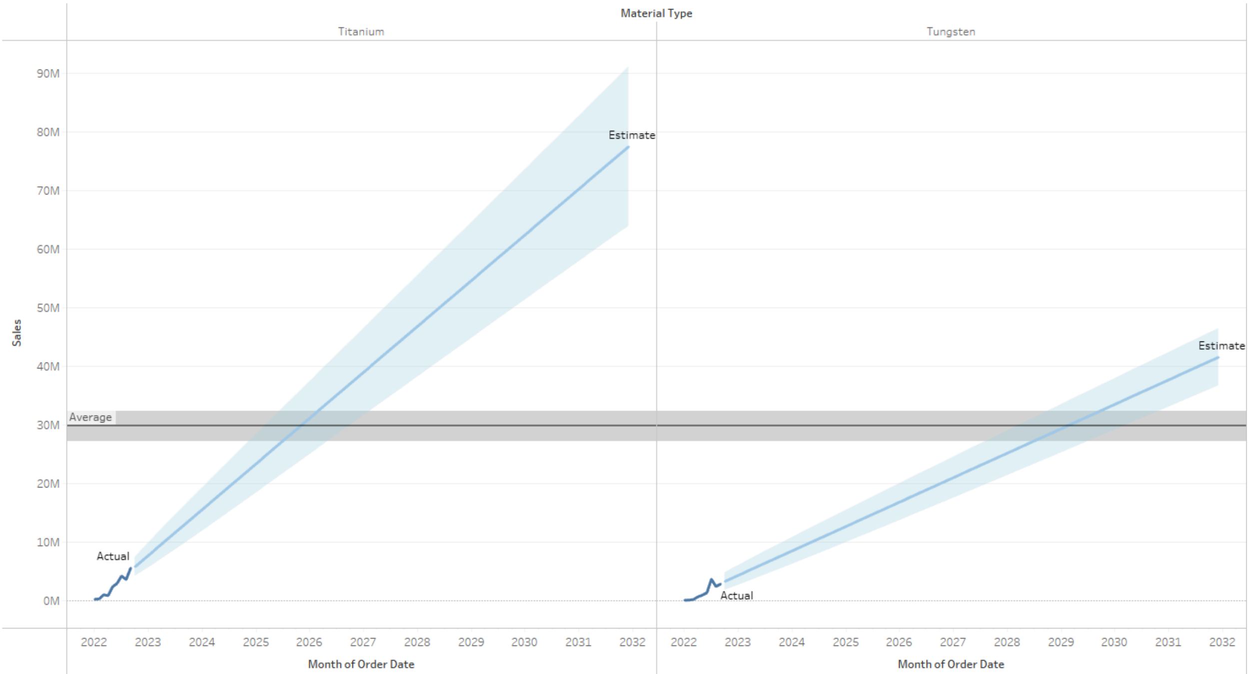
Options Used to Create Forecasts

Time series: Month of Order Date
Measures: Sum of Sales
Forecast forward: 111 months (October 2022 – December 2031)
Forecast based on: January 2022 – September 2022
Ignore last: 1 month (October 2022)
Seasonal pattern: None (Not enough data to search for a seasonal pattern recurring every 12 Months)

Sum of Sales

Initial	Change From Initial	Seasonal Effect		Contribution		
October 2022	October 2022 – December 2031	High	Low	Trend	Season	Quality
253,415,382 ± 58,605,966	3,009,118,750	None		100.0%	0.0%	Poor

Titanium & Tungsten
10-Year Sales Forecast



Options Used to Create Forecasts

Time series: Month of Order Date
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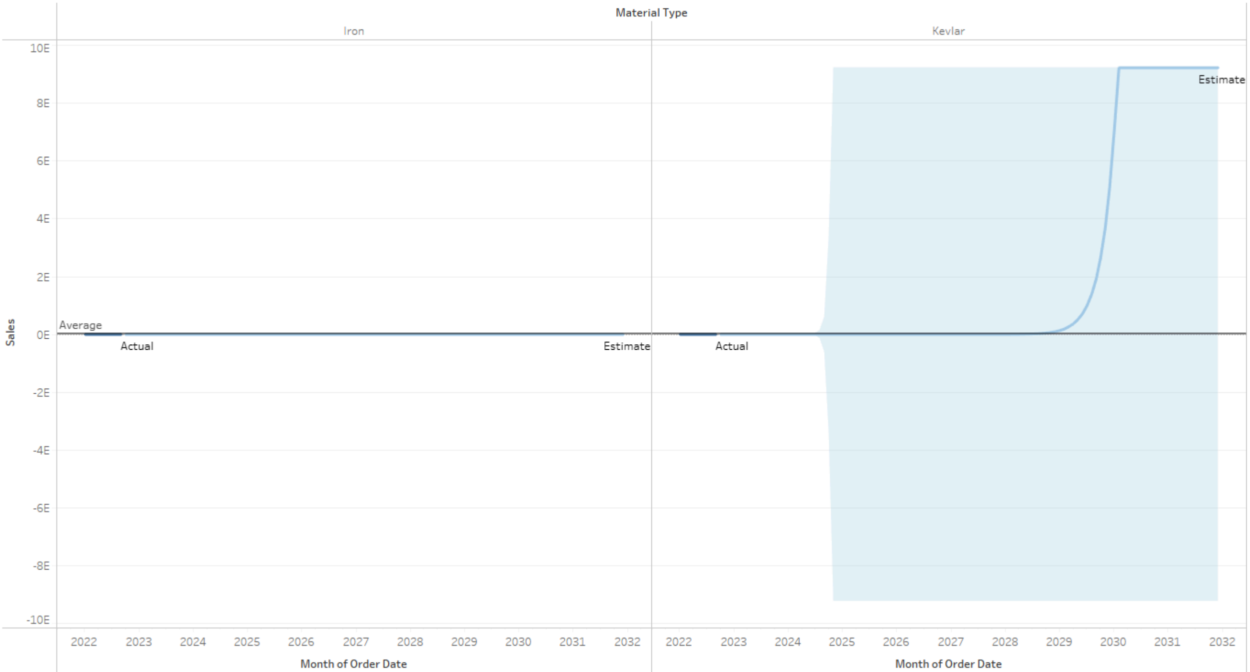
Forecast forward: 111 months (October 2022 – December 2031)
Forecast based on: January 2022 – September 2022
Ignore last: 1 month (October 2022)
Seasonal pattern: None (Not enough data to search for a seasonal pattern recurring every 12 Months)

Sum of Sales

Column	Initial	Change From Initial	Seasonal Effect		Contribution		
Material Type	October 2022	October 2022 – December 2031	High	Low	Trend	Season	Quality
Titanium	5,833,710 ± 1,575,232	71,700,930	None		100.0%	0.0%	Poor
Tungsten	3,329,549 ± 1,483,440	38,259,375	None		100.0%	0.0%	Poor

Iron vs. Kevlar
10-Year Sales Forecast



Options Used to Create Forecasts

Time series:	Month of Order Date
Measures:	Sum of Sales
Forecast forward:	111 months (October 2022 – December 2031)
Forecast based on:	January 2022 – September 2022
Ignore last:	1 month (October 2022)
Seasonal pattern:	None (Not enough data to search for a seasonal pattern recurring every 12 Months)

Sum of Sales

Column	Initial	Change From Initial	Seasonal Effect		Contribution		
Material Type	October 2022	October 2022 – December 2031	High	Low	Trend	Season	Quality
Iron	12,562,349 ± 3,526,603	143,385,000	None		100.0%	0.0%	Poor
Kevlar	3,139,148 ± 3,075,544	13,075,384,081,560,900,000,000	None		100.0%	0.0%	Ok

Appendix A

- SQL Code Used to Clean and Update Data using MySQL database (3 txt.files)

```
*********************** Reconfiguring Costs***********************

use cave_llc;
select * from orderlist;

UPDATE orderlist SET Retail = 1600.00 WHERE PlantCode = 'PLANT03';
UPDATE orderlist SET Retail = 2200.00 WHERE PlantCode = 'PLANT12';
UPDATE orderlist SET Retail = 2000.00 WHERE PlantCode = 'Plant16';
UPDATE orderlist SET Retail = 1750.00 WHERE PlantCode = 'PLANT08';
UPDATE orderlist SET Retail = 1500.00 WHERE PlantCode = 'PLANT13';
UPDATE orderlist SET Retail = 1300.00 WHERE PlantCode = 'PLANT09';
UPDATE orderlist SET Retail = 1000.00 WHERE PlantCode = 'PLANT04';

-------- Determine OrderID count by Plant----------------------
SELECT count(OrderID) from orderlist where PlantCode = 'Plant16';
-- 150 Titanium
SELECT count(OrderID) from orderlist where PlantCode = 'Plant03';
-- 6432 Steel
SELECT count(OrderID) from orderlist where PlantCode = 'Plant08';
-- 83 Tungsten
SELECT count(OrderID) from orderlist where PlantCode = 'Plant04';
-- 0
SELECT count(OrderID) from orderlist where PlantCode = 'Plant12';
-- 239 Iron
SELECT count(OrderID) from orderlist where PlantCode = 'Plant09';
-- 2 Iron
SELECT count(OrderID) from orderlist where PlantCode = 'Plant13';
-- 65
select * from orderlist;
SELECT COUNT(Sales) from orderlist WHERE Sales < 5;
SELECT MIN(Sales) FROM Orderlist;
SELECT SUM(Sales) FROM Orderlist WHERE Sales <18000;

ALTER TABLE orderlist DROP COLUMN SALES;
ALTER TABLE orderlist DROP COLUMN Cost;

------------------------- Add Sales where Retail * Quantity
ALTER TABLE orderlist ADD COLUMN Sales int(25);
ALTER TABLE orderlist DROP COLUMN Sales;
ALTER TABLE orderlist ADD COLUMN Sales INT GENERATED ALWAYS AS (Retail *
UnitQuantity) STORED;
```

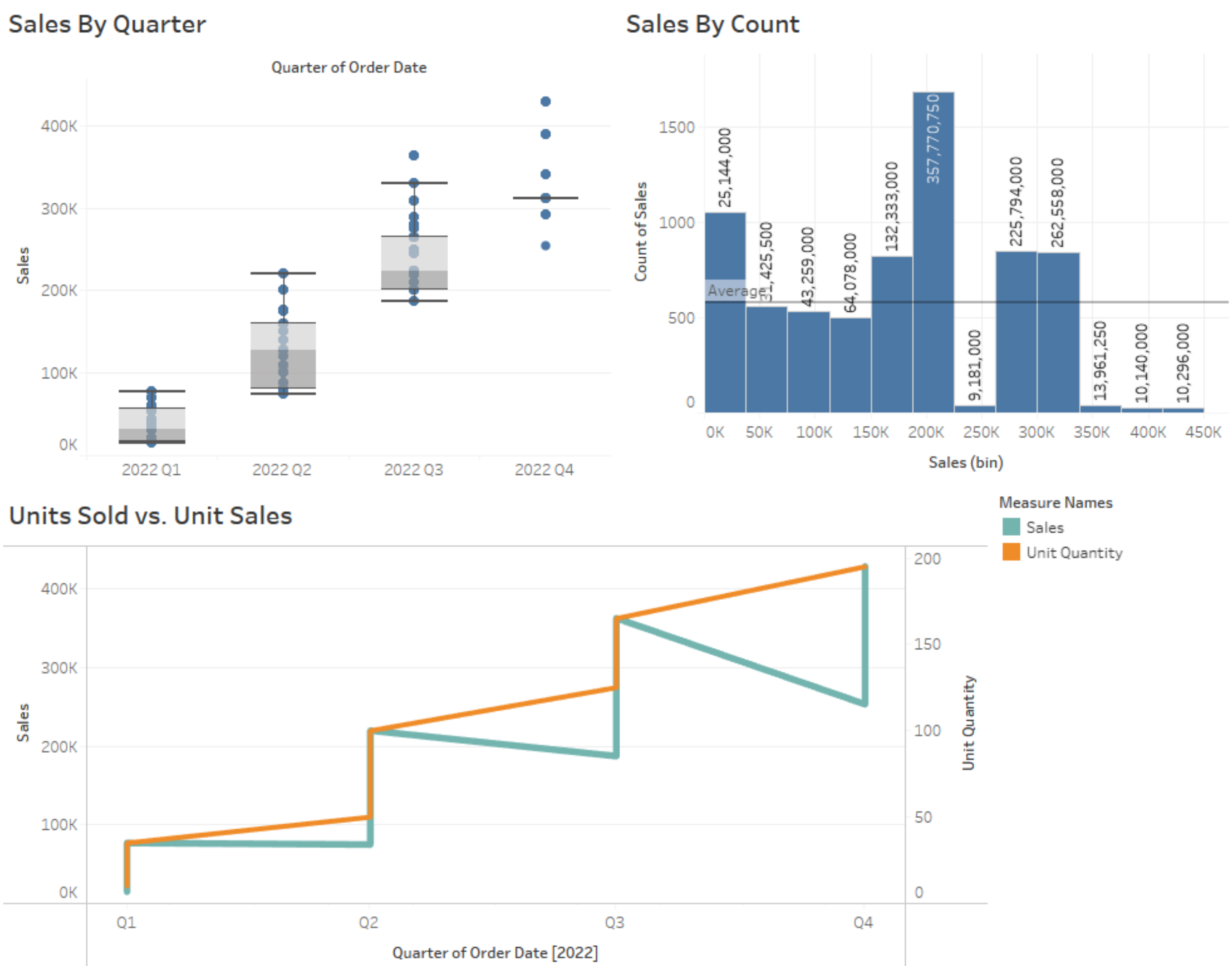
```
---------------------------- Adjust Sales Data
SELECT Sales FROM orderlist ORDER BY Sales DESC;
UPDATE orderlist SET UnitQuantity = 28 WHERE UnitQuantity = 8;
SELECT distinct Count(sales) from orderlist;

**************************** Updating Data *******************************
USE cave_llc;
SELECT * FROM orderlist;
select Distinct ProductID from orderlist order by ProductID desc;
SELECT COUNT(Distinct ProductID) from orderlist;
SELECT DISTINCT PlantCode FROM orderlist;
-- --------------SELECT COUNT(PlantCode) from orderlist WHERE PlantCode =
'PLANT13';
UPDATE orderlist SET ProductID = 1702654 WHERE PlantCode = 'PLANT16';
UPDATE orderlist SET ProductID = 1702653 WHERE PlantCode = 'PLANT03';
UPDATE orderlist SET ProductID = 1702652 WHERE PlantCode = 'PLANT08';
UPDATE orderlist SET ProductID = 1702224 WHERE PlantCode = 'PLANT04';
UPDATE orderlist SET ProductID = 1700980 WHERE PlantCode = 'PLANT12';
UPDATE orderlist SET ProductID = 1700738 WHERE PlantCode = 'PLANT09';
UPDATE orderlist SET ProductID = 1700875 WHERE PlantCode = 'PLANT13';
------------------------ ADD PRODUCT MATERIAL COLUMN AND DATA
ALTER TABLE orderlist ADD COLUMN Material char(25);
UPDATE orderlist SET Material = 'Titanium' WHERE PlantCode = 'Plant16';
UPDATE orderlist SET Material = 'Steel' WHERE PlantCode = 'PLANT03';
UPDATE orderlist SET Material = 'Tungsten' WHERE PlantCode = 'PLANT08';
UPDATE orderlist SET Material = 'Aluminum' WHERE PlantCode = 'PLANT04';
UPDATE orderlist SET Material = 'Iron' WHERE PlantCode = 'PLANT12';
UPDATE orderlist SET Material = 'Polyethylene' WHERE PlantCode = 'PLANT09';
UPDATE orderlist SET Material = 'Kevlar' WHERE PlantCode = 'PLANT13';
---------------------------- ADD COST COLUMN AND ADD PRICING DATA
ALTER TABLE orderlist ADD COLUMN cost int(15);
UPDATE orderlist SET cost = 15000 WHERE PlantCode = 'Plant16';
UPDATE orderlist SET cost = 25000 WHERE PlantCode = 'PLANT03';
UPDATE orderlist SET cost = 23000 WHERE PlantCode = 'PLANT08';
UPDATE orderlist SET cost = 2000 WHERE PlantCode = 'PLANT04';
UPDATE orderlist SET cost = 17000 WHERE PlantCode = 'PLANT12';
UPDATE orderlist SET cost = 5000 WHERE PlantCode = 'PLANT09';
UPDATE orderlist SET cost = 20000 WHERE PlantCode = 'PLANT13';
------------------------- Add Cost*Weight
ALTER TABLE orderlist ADD COLUMN Sales int(25);
ALTER TABLE orderlist DROP COLUMN Sales;
ALTER TABLE orderlist ADD COLUMN Sales INT GENERATED ALWAYS AS (Weight *
Cost) STORED;
```

```
-----------------------------------------------------------------------------------------------------

SELECT * FROM orderlist WHERE OrderID LIKE '144715%'; -- 384 ROWS

DELETE FROM orderlist WHERE OrderDate > '2021-07-01';
INSERT INTO orderlist SET OrderDate = 2020-01-05 WHERE OrderID LIKE '1447%0';
INSERT INTO orderlist(OrderDate) VALUES ('2020-01-05');

SELECT * FROM orderlist WHERE OrderDate = '2020-01-05';
SELECT OrderID FROM orderlist ORDER BY OrderID ASC;
ALTER TABLE orderlist DROP COLUMN OrderDate;
DELETE FROM orderlist WHERE OrderDate LIKE '14471%';
SELECT * FROM orderlist;
SELECT OrderID from orderlist order by OrderID asc;
SELECT DISTINCT(ORDERID) FROM orderlist order by OrderID asc;

*********************** ORDER DATES/UPDATE UNITQTY ********************
SELECT * FROM orderlist;
SELECT * FROM orderlist WHERE OrderDate between '2022-01-01' AND '2022-01-30';
-- 1000 rows or more
SELECT * FROM orderlist WHERE OrderDate between '2022-04-01' AND '2022-06-30';
-- 1000 rows
SELECT * FROM orderlist WHERE OrderDate between '2022-07-01' AND '2022-09-30';
-- 455 rows
SELECT * FROM orderlist WHERE OrderDate between '2022-10-01' AND '2022-12-30';
-- 521 rows
SELECT * FROM orderlist WHERE OrderDate between '2022-01-01' AND '2022-12-30';
-- 138 rows
SELECT * FROM orderlist WHERE OrderDate NOT between '2022-01-01' AND
'2022-12-30'; -- 219 rows
SELECT * FROM orderlist WHERE OrderDate NOT between '2022-10-02' AND
'2022-12-30'; -- 138 rows


SELECT COUNT(DISTINCT OrderID) from orderlist;
SELECT COUNT(DISTINCT UnitQuantity) from orderlist;  -- 16 Distinct Types;
-- correction

UPDATE orderlist SET OrderDate = '2022-01-12' WHERE OrderID LIKE '1447%1';
UPDATE orderlist SET UnitQuantity = '10' WHERE OrderID LIKE '1447%1';
SELECT * FROM orderlist WHERE OrderID LIKE '1447%1';  -- 548 rows
SELECT SUM(Sales) FROM Orderlist WHERE OrderDate = '2022-01-12'; -- 13,475,000

UPDATE orderlist SET OrderDate = '2022-02-22' WHERE OrderID LIKE '1447%2';
```

```
UPDATE orderlist SET UnitQuantity = '20' WHERE OrderID LIKE '1447%2';
SELECT * FROM orderlist WHERE OrderID LIKE '1447%2'; -- 527 rows
SELECT SUM(Sales) FROM Orderlist WHERE OrderDate = '2022-02-22'; -- 25,992,000

UPDATE orderlist SET OrderDate = '2022-03-12' WHERE OrderID LIKE '1447%3';
UPDATE orderlist SET UnitQuantity = '35' WHERE OrderID LIKE '1447%3';
SELECT * FROM orderlist WHERE OrderID LIKE '1447%3'; -- 547 rows
SELECT SUM(Sales) FROM Orderlist WHERE OrderDate = '2022-03-12'; -- 47,239,500

UPDATE orderlist SET OrderDate = '2022-04-22' WHERE OrderID LIKE '1447%4';
UPDATE orderlist SET UnitQuantity = '50' WHERE OrderID LIKE '1447%4';
SELECT * FROM orderlist WHERE OrderID LIKE '1447%4'; -- 523 rows
SELECT SUM(Sales) FROM Orderlist WHERE OrderDate = '2022-04-22'; -- 64,280,000

UPDATE orderlist SET OrderDate = '2022-05-02' WHERE OrderID LIKE '1447%5';
UPDATE orderlist SET UnitQuantity = '80' WHERE OrderID LIKE '1447%5';
SELECT * FROM orderlist WHERE OrderID LIKE '1447%5';  -- 531 rows
SELECT SUM(Sales) FROM Orderlist WHERE OrderDate = '2022-05-02'; --
104,240,000

UPDATE orderlist SET OrderDate = '2022-06-22' WHERE OrderID LIKE '1447%0';
UPDATE orderlist SET UnitQuantity = '100' WHERE OrderID LIKE '1447%0';
SELECT * FROM orderlist WHERE OrderID LIKE '1447%0'; -- 826 rows
SELECT SUM(Sales) FROM Orderlist WHERE OrderDate = '2022-06-22'; --
203,740,000

UPDATE orderlist SET OrderDate = '2022-07-30' WHERE OrderID LIKE '1447%6';
UPDATE orderlist SET UnitQuantity = '125' WHERE OrderID LIKE '1447%6';
SELECT * FROM orderlist WHERE OrderID LIKE '1447%6';  -- 870 rows
SELECT SUM(Sales) FROM Orderlist WHERE OrderDate = '2022-07-30'; --
267,150,000

UPDATE orderlist SET OrderDate = '2022-08-12' WHERE OrderID LIKE '1447%7';
UPDATE orderlist SET UnitQuantity = '140' WHERE OrderID LIKE '1447%7';
SELECT * FROM orderlist WHERE OrderID LIKE '1447%7'; -- 894 rows
SELECT SUM(Sales) FROM Orderlist WHERE OrderDate = '2022-08-12'; --
308,420,000

UPDATE orderlist SET OrderDate = '2022-09-22' WHERE OrderID LIKE '1447%8';
UPDATE orderlist SET UnitQuantity = '165' WHERE OrderID LIKE '1447%8';
SELECT * FROM orderlist WHERE OrderID LIKE '1447%8'; -- 847 rows
SELECT SUM(Sales) FROM Orderlist WHERE OrderDate = '2022-09-22'; --
343,860,000
```

```
UPDATE orderlist SET OrderDate = '2022-10-22' WHERE OrderID LIKE '1447%9';
UPDATE orderlist SET UnitQuantity = '195' WHERE OrderID LIKE '1447%9';
SELECT * FROM orderlist WHERE OrderID LIKE '1447%9'; -- 858 rows
SELECT SUM(Sales) FROM Orderlist WHERE OrderDate = '2022-10-22'; --
411,118,500
```

Appendix B - Tableau Dashboard Webpages



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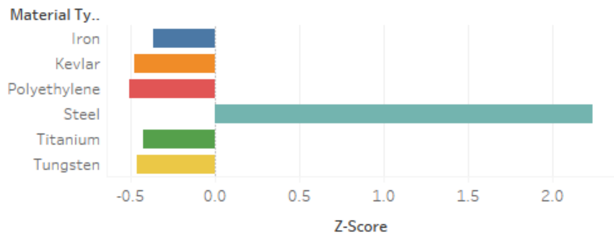
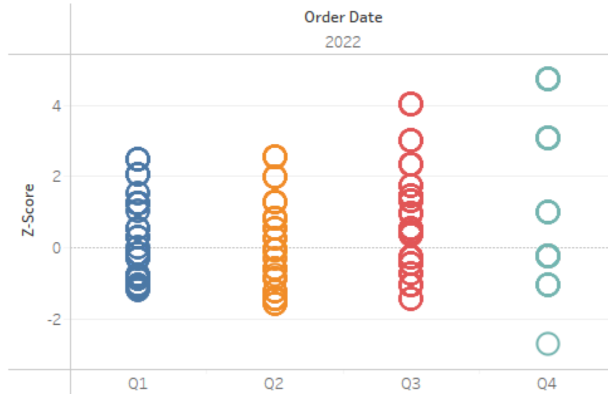
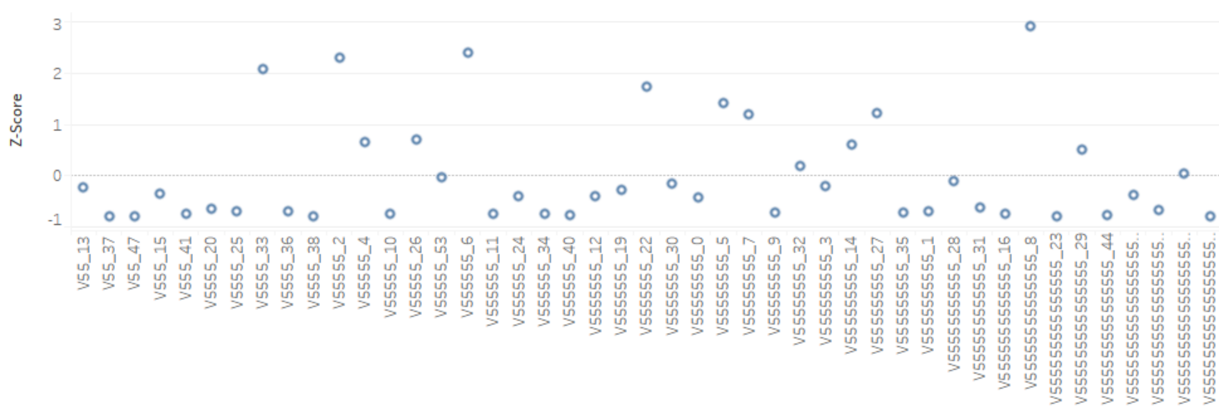
Z-Score by Material Type



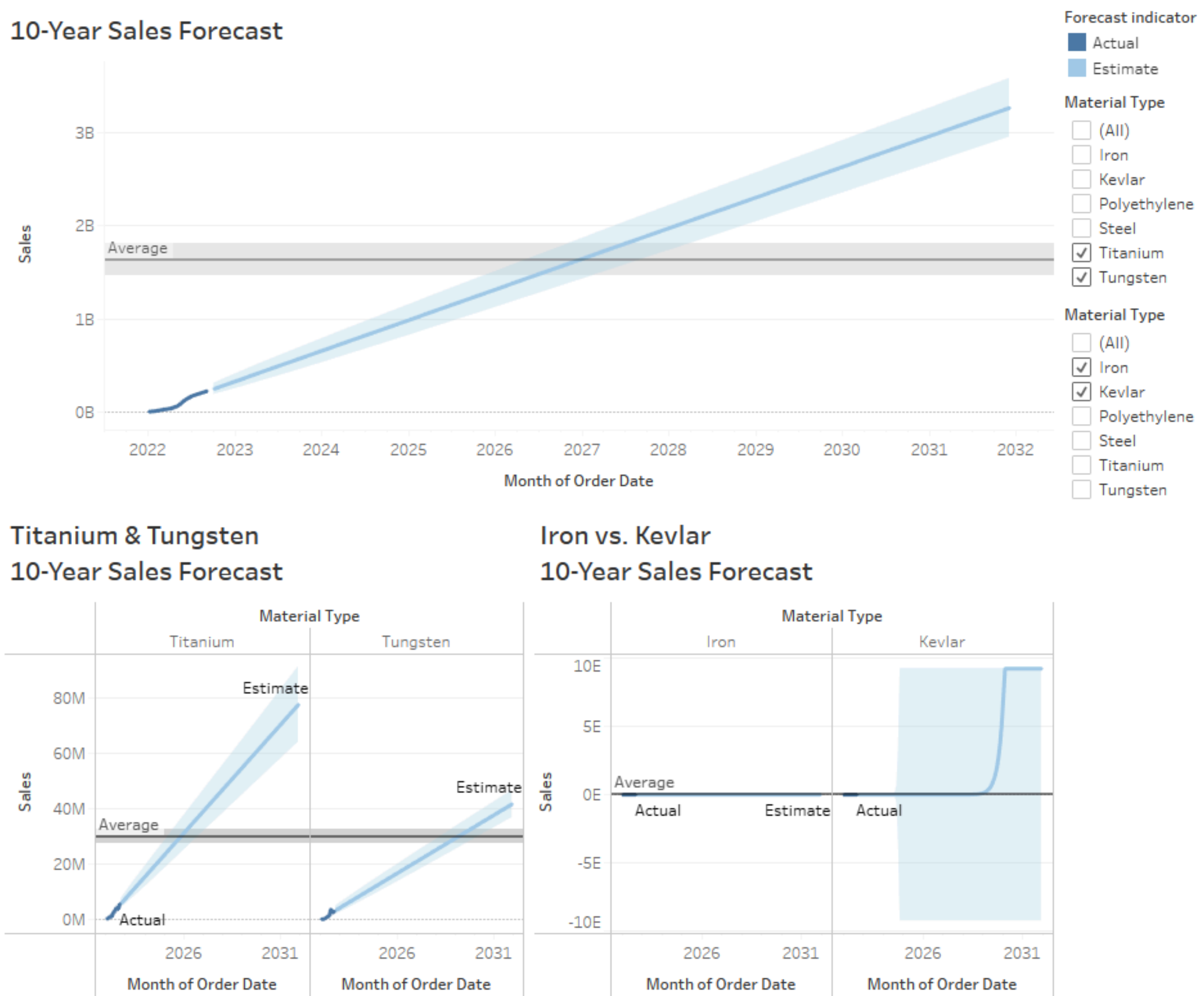
Z-Score by Frequency of Orders by Quarter



Z-Score by Sales by Customer



https://public.tableau.com/views/CaveLLCZScoreAnalysis/ZscoreDash?:language=en-US&publish=yes&:display_count=n&:origin=viz_share_link



https://public.tableau.com/views/CaveLLCForecastAnalysis/10YrForecastDashboard?:language=en-US&publish=yes&:display_count=n&:origin=viz_share_link

APPENDIX C

- Link to Google Site
  - Link to usable form in Google Site